

M110 plus 01

Development Document

Version V1.1

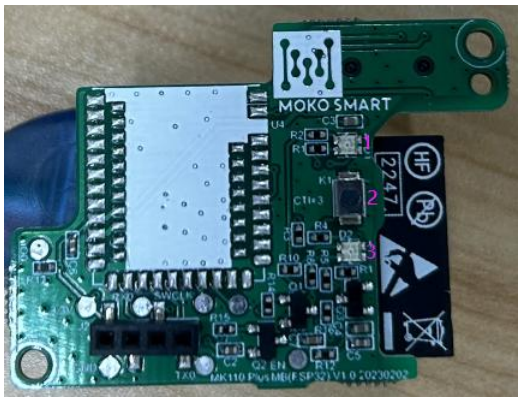
This document is provided for customers to develop their own firmware and flash the custom firmware on MK110 plus 01 hardware.

1. Hardware Description

The product MK110 plus 01 has two board: the power board, the control board. Firmware download can be done using only the control board.

Top View

control board

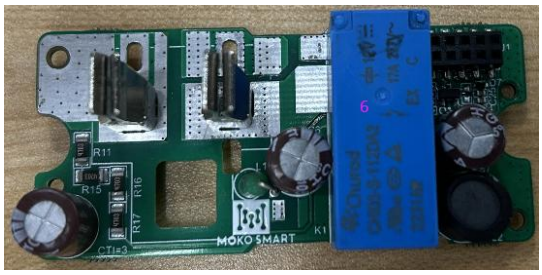


Bottom View

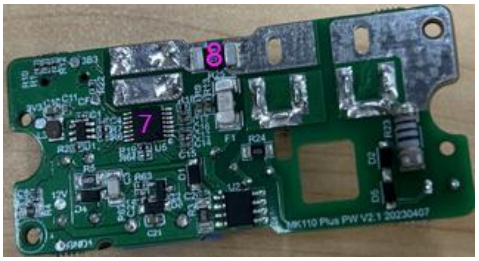
control board



power board



power board



Assembled



Figure 1 Hardware Photos of MK110 plus 01

No.	Location	Name	Description
1	D1	LED1	1615RGB

2	K1(control board)	BUTTON	/
3	D2	LED2	1615RGB
4	U3	Control module	Wifi &BLE module (ESP32-WROOM-32E)
5	U1	EEPROM	BL24C64A
6	K1(power board)	RELAY	Normal open relay
7	U5	Single-phase Electrical Energy Meter IC	BL0942
8	R25	Resistor for current measure	2mR

2. Schematic Diagram of the Control Board

The WIFI &BLE module is ESP32-WROOM-32E, it is located on the control board U3. The schematic and pins definition are as below.

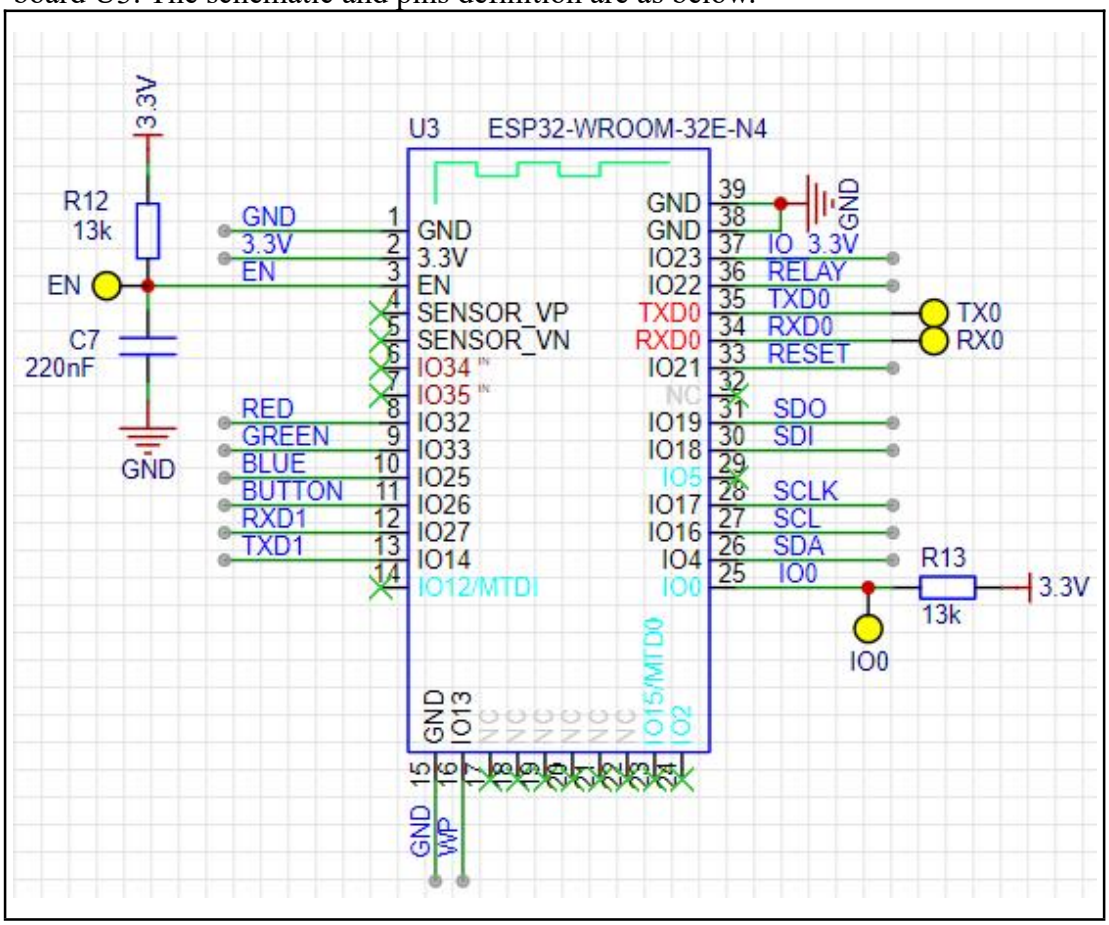
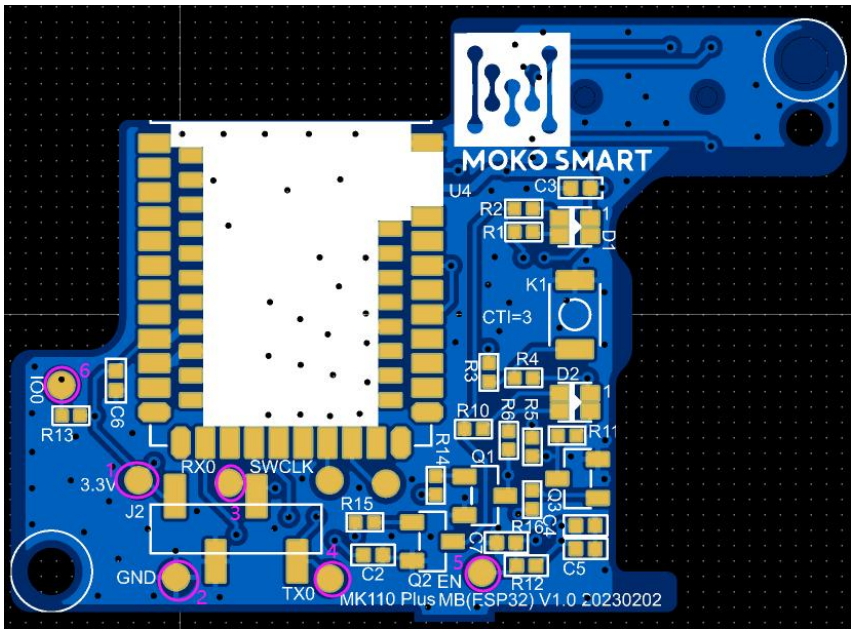


Figure 2 Schematic diagram of the control module

Pin number of U3	Pin of ESP32	Function	Notes
1&15&38&39	GND	GND	GND
2	3.3V	3.3V	3.3V
3	EN	EN	reset pin, has reserved RC externally
8	IO32	RED	This pin is high, red on D1 and D2
9	IO33	GREEN	This pin is high, green on D1 and D2
10	IO25	BLUE	This pin is high, blue on D1 and D2
11	IO26	BUTTON	It's pulled up to 3.3V. When press down the button, the input signal on the pin is low level. It is labeled with K1.
12&13	IO27&IO14	RXD1&TXD1	Unused
16	IO13	WP	Write protect for BL24C64A.
26	IO4	SDA	The EEPROM is BL24C64A and supplied by 3.3V. It is labeled with U1.
27	IO16	SCL	
25	IO0	IO0	IO0 is used to download firmware. Please see figure 3 to get the wiring method
28	IO17	SCLK	They are used for SPI communication with BL0942
30	IO18	SDI	
31	IO19	SDO	
33	IO21	RESET	Unused
34	RXD0	RXD0	RXD0 is used to download and debug. Please see figure 3 to get the wiring method.
35	TXD0	TXD0	TXD0 is used to download and debug. Please see figure 3 to get the wiring method.
36	IO22	RELAY	It is a normal-open relay. If the pin output high level, the normal-open relay will be closed.

37	IO23	IO_3.3V	This pin should be low. When this pin is on high power, the control board will have no 3.3V power supply

3. Test Point Description



top side of the Control board

Figure 3 the test points on the control board

Test points in figure 3 are related to ESP32 firmware download.

NO.	Test points on the board	Description
1	3.3V	Link to the 3.3V of USB to Serial
2	GND	Link to the GND of USB to Serial
3	RX0	Link to the TXD of USB to Serial
4	TX0	Link to the RXD of USB to Serial
5	EN	Link to the RTS of USB to Serial
6	IO0	Link to the GND of USB to Serial

4. Revision History

Revision	Description	Editor	Date
V1.0	Initial Version	TSX	2023.6.15
V1.1	Added interface information about the metering chip	TSX	2023.9.26

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